

Syed Mahdi Rehan

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SKILLS OVERVIEW

- **Engineering Software tools** – ArcGIS, 3D SolidWorks, AutoCAD, MATLAB, Microsoft Word, Excel, CVAT.
- **Computer Languages** – Python, Java, C++.

EDUCATION

Bachelor of Engineering | Engineering Systems & Computing (Co-op)

Sept 2022 - Present

University of Guelph, Guelph, ON

- Relevant courses: Engineering Design, Mechatronics Systems Design, Material Science, Engineering Analysis.

WORK EXPERIENCE

AI Engineer Intern

May 2025 – Sept 2025

Engineering Consultants International Limited, Dubai, UAE

- Developed YOLOv8 segmentation model (18,000+ images, 81.5% accuracy) to detect and classify 10 pavement distress types by severity, including real-world dimension extraction.
- Automated Pavement Condition Index (PCI) calculation using ASTM D6433 standards to evaluate pavement condition, integrating outputs with interactive GIS maps and Excel data organization templates.
- Collaborated on designing and developing a desktop application using React.js and Python for pavement survey data processing, automated distress detection, and PCI reporting to provide data driven maintenance decision.

Mobile Application Developer Intern

Dec 2024 – Feb 2025

AMH Tech Solutions, London, Ontario, Canada.

- Developed an Android app in Java to capture GPS, accelerometer, and video data, and compute IRI (International Roughness Index) from smartphone metadata for automated road quality assessment.
- Validated results against LCMS benchmarks, achieving 90% accuracy, enabling cost-effective and portable pavement quality assessment.

Engineering Co-Op Work Term

June 2024 – Sept 2024

Engineering Consultants International Limited, Dubai, UAE

- Developed Operational Asset Management Plans (OAMPs) for 28 road assets, ensuring regulatory compliance.
- Gained hands-on experience with advanced data collection technologies (LiDAR, GPR, LCMS) and data integration using ArcGIS for data analysis and visualization.
- Utilized ROMDAS and data analytics to evaluate road infrastructure condition and develop optimized maintenance plans, improving asset longevity, public safety and cost efficiency.

ENGINEERING PROJECTS

Traffic Flow Model Simulation in Python

March 2025 – April 2025

University of Guelph, Guelph, ON

- Developed a Python-based Cellular Automata traffic simulation to model real-world vehicle behaviors including acceleration, braking, lane changing, merging and traffic signal interactions.
- Enhanced the model with innovative dynamic signals and slow zones, and implemented customizable parameters (vehicle density, lane count, speed limits, signal intervals) to simulate diverse traffic scenarios.
- Generated congestion heatmaps, flow-density plots and animations to evaluate traffic flow based on the given input parameters, and to demonstrate how adaptive dynamic signals improved overall traffic flow efficiency.

CERTIFICATIONS

Certified SolidWorks Associate – Mechanical Design

March 2024

University of Guelph, Guelph, ON